

GEE Runbook — Tables S4 and S7

RiceBaCI v2.0 reviewer rebuttal · pre-submission populate

This runbook walks you click-by-click through two Google Earth Engine sessions that produce the real numbers for two currently-placeholder supplement tables:

- **Table S4** — Cell-level Difference-in-Differences (DiD) with cell and year fixed effects
- **Table S7** — Cell-level placebo / pre-trend test

After you finish both sessions and upload the eight resulting CSVs back to this thread, the assistant will run Module 14 and Module 15 to produce the final tau-hat, standard errors, and confidence intervals, then rebuild Supplement_Combined.docx with the real numbers spliced into Tables S4 and S7.

Prerequisites

Item	Value
GEE Cloud project	durable-pulsar-486209-b5
GEE account	the same Google account used for Modules 04 / 04b earlier
Browser	Chrome or Edge (latest)
Free Drive space	≥ 500 MB
Runtime — Module 13	~ 1.5 hours (8 districts; 8–20 min each, run in parallel)
Runtime — Module 04b	~ 3 hours (8 districts; 15–30 min each, run in parallel)

*Note: both sessions are **fully deterministic** — run them once now and the same numbers will reproduce on demand for the editor or any reviewer.*

Session 1 — Module 13 (Landsat pre-trends, 2014–2018)

Goal

Produce a 5-year pre-treatment phenology panel from harmonised Landsat 5/7/8 for the same 8 districts and same Kharif window used in the main panel. The output feeds Module 14 (event-study + placebo) → Table S2 will be re-verified and the joint pre-trend Wald test in §S4 will be re-run on real numbers.

Steps

- 1 Open <https://code.earthengine.google.com> and sign in with the Google account that has the **durable-pulsar-486209-b5** Cloud project enabled.
- 2 Top-right gear icon → **Project** → confirm the active project is **durable-pulsar-486209-b5**. If not, switch to it.
- 3 Click **+ New** → **File**. Name the new script **module_13_landsat_pretrends**.
- 4 Open the file `gee_module_13_landsat_pretrends.js` from this thread on your computer. Select all (Ctrl/Cmd-A), copy, and paste into the new GEE script tab.
- 5 Click **Save**, then click **Run**. Wait for the console message: *“Module 13 ready. Open the Tasks tab → click Run on each of the 8 export tasks.”*
- 6 Switch to the **Tasks** tab (right panel). You should see **8 unstarted tasks** named `bacl_panel_landsat_<District>`.
- 7 Click **Run** next to each task. A dialog will pop up — accept the defaults and click **Run** again. (You can fire all 8 in parallel — GEE will queue them.)
- 8 Each task runs 8–20 minutes. When each finishes, it writes one CSV to your Google Drive folder **RiceBaCI_landsat_pretrends** (auto-created).
- 9 Once all 8 CSVs are present in Drive: open Drive, select the 8 files, right-click → **Download**. Drive will zip them.
- 10 Unzip locally. You should have 8 files named `bacl_panel_landsat_<District>.csv`.
- 11 **Concatenate them into a single file** using any of the methods on page 4 (Concatenation cheat-sheet). Name the merged file **bacl_panel_landsat_2014_2018.csv**.
- 12 Upload **bacl_panel_landsat_2014_2018.csv** back to this Perplexity thread and reply: *“Session 1 done, file attached.”*

Session 2 — Module 04b (Balanced-pixel panel, 2019–2024)

Goal

Export a cell-level (100 m grid cell) SOS/POS/EOS panel restricted to pixels that pass quality control in **every** year 2019–2024. This balanced sub-sample is what Model 2 (Table S4 cell-level DiD) and the cell-level placebo (Table S7) require. The script draws up to 1,000 random cells per district (seed = 42, deterministic) — total panel size $\approx 8 \times 1000 \times 6 \times 3 \approx 144$ k rows.

Steps

- 1 In the same GEE code editor, click **+ New** → **File**. Name it **module_04b_balanced_pixel**.
- 2 Open `gee_module_04b_balanced_pixel.js` from this thread, select all, copy, and paste into the new script tab.
- 3 Click **Save**, then click **Run**. Wait for the console message: *“Module 04b ready. Open the Tasks tab → click Run on each of the 8 export tasks.”*
- 4 Switch to the **Tasks** tab. You should see **8 new unstarted tasks** named `bacl_balanced_<District>`.
- 5 Click **Run** next to each task (accept defaults). These tasks are slower than Module 13 — expect 15–30 minutes each. Run them in parallel.
- 6 When complete, the 8 CSVs land in Google Drive folder **RiceBaCI_balanced_pixel**.
- 7 Download all 8 from Drive, unzip, and concatenate them into a single file **bacl_panel_balanced_pixel.csv** (see cheat-sheet on next page).
- 8 Upload **bacl_panel_balanced_pixel.csv** back to this Perplexity thread and reply: *“Session 2 done, file attached.”*

What to watch out for in Module 04b: if any district task fails with *“User memory limit exceeded”*, lower `N_PER_DISTRICT` from 1000 to 500 (line 28 of the script) and re-run just that district's task. The downstream DiD has plenty of statistical power even at $500 \text{ cells} \times 6 \text{ years} = 3,000 \text{ obs}$.

Concatenation cheat-sheet

Pick the option that matches your local tooling.

Option A — Python (recommended)

In a terminal in the folder where you unzipped the 8 CSVs:

```
python -c "import pandas as pd, glob; pd.concat([pd.read_csv(f) for f in
sorted(glob.glob('baclI_*.csv'))]).to_csv('MERGED.csv', index=False)"
```

Rename MERGED.csv to either bacI_panel_landsat_2014_2018.csv (Session 1) or bacI_panel_balanced_pixel.csv (Session 2).

Option B — Shell (Linux / macOS / Git Bash)

```
head -n 1 bacI_panel_landsat_Baleshwar.csv > MERGED.csv && \ for f in
bacI_panel_landsat_*.csv; do tail -n +2 "$f" >> MERGED.csv; done
```

Option C — Excel / LibreOffice

Open the first CSV. Open each of the remaining 7 CSVs, copy all rows **except the header**, and paste at the bottom of the first file. Save As → CSV (UTF-8), with the merged filename above.

What happens after you upload

- 1 The assistant runs **Module 14** on bacI_panel_landsat_2014_2018.csv + the existing bacI_panel_real.csv → produces table_S10_pretrend_event_study.csv and table_S11_placebo_did.csv, plus an updated joint Wald test on parallel trends.
- 2 The assistant runs **Module 15** on bacI_panel_balanced_pixel.csv → produces table_S4_cell_level_did.csv (Model 2 tau-hat for SOS, POS, EOS) and table_S7_cell_level_placebo.csv (cell-level pre-trend placebo).
- 3 The assistant rebuilds **Supplement_Combined.docx**: §S4 tables S4 and S7 will be re-rendered with the new tau-hat, SE, 95 % CI, p-values, and interpretation strings; §S4 prose around them will be updated to reference the real numbers; the joint pre-trend Wald test result in §S4 will be patched in.
- 4 The assistant shares the new **Supplement_Combined.docx** back to you (same shared-asset name, version 2 in history).

If anything breaks — paste the GEE error message verbatim into the thread. Common issues: (a) Cloud project not set → switch project; (b) task fails with "User memory limit exceeded" → lower N_PER_DISTRICT; (c) one or two districts return zero balanced cells → that's expected in extreme cloud-cover years; we'll handle it in Module 15 (rows are simply dropped from the corresponding district).

Files in this thread (re-download fresh each session)

File	Purpose
gee_module_13_landsat_pretrends.js	GEE script — Session 1
gee_module_04b_balanced_pixel.js	GEE script — Session 2

module_14_pretrend_placebo.py	Python consumer — Tables S2 & joint Wald
module_15_balanced_pixel_did.py	Python consumer — Tables S4 & S7
GEE_Runbook_Tables_S4_S7.pdf	This runbook

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Repository: github.com/pandasupranab/RiceBaCI-GEE (branch v2.0-refit)

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